EXP.NO: 12 Develop a JavaScript-based validation mechanism for a Library Management System and integrate it with a Servlet/JSP-based backend for database access

AIM:

To design a client-side validation system using JavaScript for forms in a Library Management System and ensure proper data handling before sending it to a Servlet/JSP-based backend connected to a database.

ALGORITHM:

1. Input:

 The user interacts with various forms such as:

o User Registration Form

o Book Entry Form

o Book Issue/Return Form

2. Steps for Form Validation Using JavaScript:

1. Identify Forms and Fields:

 Registration Form:

o Name, Email, Phone Number, Membership ID

 Book Form:

o Book ID, Title, Author, ISBN

 Issue Form:

o User ID, Book ID, Issue Date, Return Date

2. Apply Validation Rules (Client-Side):

 Name: Should not be empty, only alphabets allowed

 Email: Must follow standard email format

 Phone Number: Must be exactly 10 digits

 Membership ID / Book ID: Should not be empty and must follow a specific format

 ISBN: Must be numeric and valid length

 Dates:

o Issue date cannot be in the future

o Return date must be after issue date

3. Implement Validation Logic:

 When the user submits a form:

o JavaScript checks each field based on rules

o If invalid → show error message and prevent submission

o If valid → allow form submission

4. Provide User Feedback:

 Display error messages near the fields

 Highlight incorrect inputs

 Allow correction before submission

3. Integration with Servlet/JSP Backend:

1. Form Submission:

 After successful validation, form data is sent to a JSP/Servlet using HTTP POST

2. Server-Side Processing:

 Servlet receives request parameters

 Performs additional validation (security layer)

 Interacts with database:

o Store user details

o Add/update book records

o Manage issue/return transactions

3. Database Access:

 Use JDBC in Servlet to:

o Insert records (Registration, Book Entry)

o Retrieve records (Search books, user history)

o Update records (Issue/Return status)

4. Generate Response:

 JSP page displays:

o Success messages (e.g., “Book Issued Successfully”)

o Error messages (if server validation fails)

4. Assumptions (Business Logic):

 Each user must have a unique Membership ID

 A user can borrow maximum 3 books at a time

 A book cannot be issued if no copies are available

 Return date must be greater than issue date

 Fine is applied if the book is returned after due date

 ISBN is considered unique for each book

 Only registered users can issue books

 Admin privileges are required to add or remove books

5. Deployment and Testing:

 Deploy application on a server (e.g., Apache Tomcat)

 Test different scenarios:

o Invalid inputs (empty fields, wrong formats)

o Valid inputs (successful submission)

o Database updates and retrieval

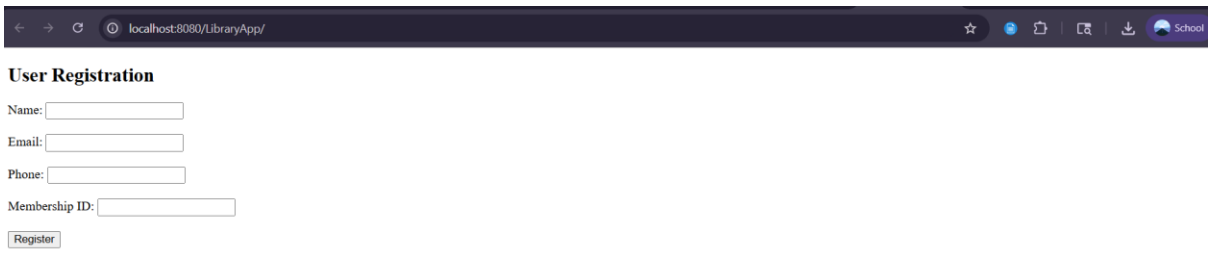
6. Observations:

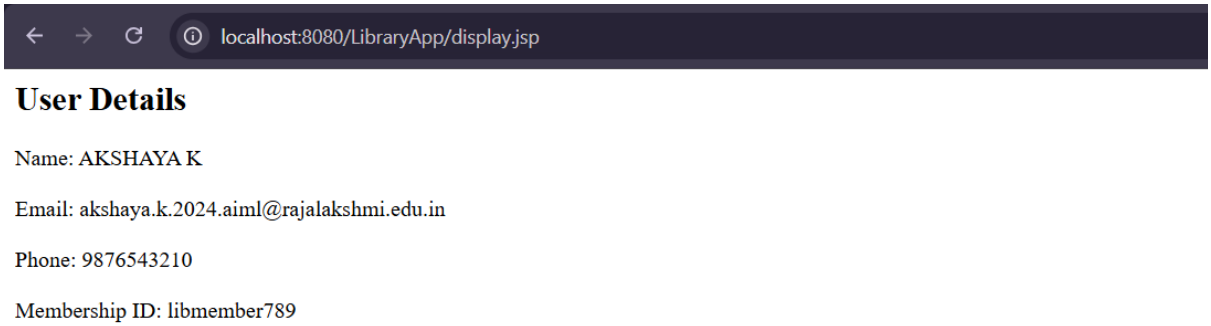
 JavaScript validation reduces invalid data submission

 Server-side validation ensures security and consistency

 Integration ensures smooth data flow between frontend and backend

OUTPUT:





RESULT:

A JavaScript-based validation system is successfully implemented for a Library Management System, ensuring correct user input before submission and seamless integration with a Servlet/JSP backend for database operations.